

TRAVAUX PRATIQUES

Administration Systeme Linux

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MODULE 01 — Gestion des Utilisateurs et des Droits

useradd · chmod · chown · setfacl · chage

Etape 1 — Creation des utilisateurs

Creation des utilisateurs alice, bob (groupe devweb) et charlie (lecture seule).

```
$ sudo useradd -m -s /bin/bash alice
```

```
$ sudo useradd -m -s /bin/bash bob
```

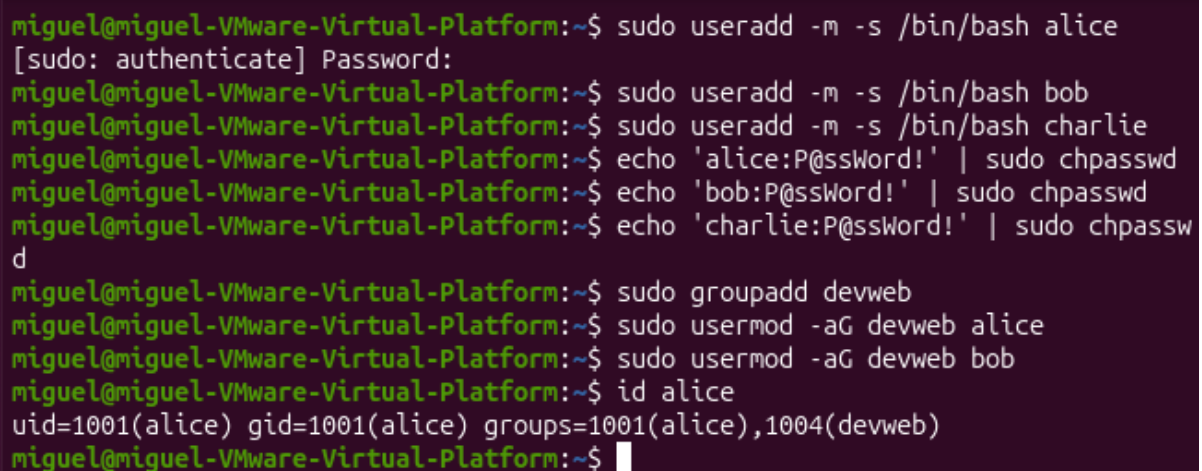
```
$ sudo useradd -m -s /bin/bash charlie
```

```
$ sudo groupadd devweb
```

```
$ sudo usermod -aG devweb alice
```

```
$ sudo usermod -aG devweb bob
```

```
$ id alice
```



```
miguel@miguel-VMware-Virtual-Platform:~$ sudo useradd -m -s /bin/bash alice
[sudo: authenticate] Password:
miguel@miguel-VMware-Virtual-Platform:~$ sudo useradd -m -s /bin/bash bob
miguel@miguel-VMware-Virtual-Platform:~$ sudo useradd -m -s /bin/bash charlie
miguel@miguel-VMware-Virtual-Platform:~$ echo 'alice:P@ssWord!' | sudo chpasswd
miguel@miguel-VMware-Virtual-Platform:~$ echo 'bob:P@ssWord!' | sudo chpasswd
miguel@miguel-VMware-Virtual-Platform:~$ echo 'charlie:P@ssWord!' | sudo chpasswd
d
miguel@miguel-VMware-Virtual-Platform:~$ sudo groupadd devweb
miguel@miguel-VMware-Virtual-Platform:~$ sudo usermod -aG devweb alice
miguel@miguel-VMware-Virtual-Platform:~$ sudo usermod -aG devweb bob
miguel@miguel-VMware-Virtual-Platform:~$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(devweb)
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 1 création des utilisateurs et id alice

Resultat attendu : id alice affiche : uid, gid et groups=devweb

Etape 2 — Repertoire et droits

```
$ sudo mkdir -p /srv/projet_web
```

```
$ sudo chown alice:devweb /srv/projet_web
```

```
$ sudo chmod 770 /srv/projet_web
```

\$ ls -la /srv/

```
miguel@miguel-VMware-Virtual-Platform:~$ sudo mkdir -p /srv/projet_web
miguel@miguel-VMware-Virtual-Platform:~$ sudo chown alice:devweb /srv/projet_web
miguel@miguel-VMware-Virtual-Platform:~$ sudo chmod 770 /srv/projet_web
miguel@miguel-VMware-Virtual-Platform:~$ ls -la /srv/
total 12
drwxr-xr-x  3 root  root   4096 Jun 10 07:45 .
drwxr-xr-x 20 root  root   4096 Jun  9 14:55 ..
drwxrwx---  2 alice devweb 4096 Jun 10 07:45 projet_web
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 2 Droits projet web

Resultat attendu : drwxrwx--- 2 alice devweb sur projet_web

Etape 3 — Configuration ACL

\$ sudo setfacl -m u:charlie:r-x /srv/projet_web

\$ getfacl /srv/projet_web

```
miguel@miguel-VMware-Virtual-Platform:~$ sudo setfacl -m u:charlie:r-x /srv/proj
et_web
miguel@miguel-VMware-Virtual-Platform:~$ getfacl /srv/projet_web
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/projet_web
# owner: alice
# group: devweb
user::rwx
user:charlie:r-x
group::rwx
mask::rwx
other::---
```

Figure 3 getfacl

Resultat attendu : getfacl affiche la ligne user:charlie:r-x

Etape 4 — Test des acces

\$ su - bob -c 'touch /srv/projet_web/test.txt'

\$ su - charlie -c 'touch /srv/projet_web/t.txt'

```

miguel@miguel-VMware-Virtual-Platform:~$ su - bob -c 'touch /srv/projet_web/test
.txt'
Mot de passe :
miguel@miguel-VMware-Virtual-Platform:~$ su - charlie -c 'touch /srv/projet_web/
test.txt'
Mot de passe :
su: Échec de l'authentification
miguel@miguel-VMware-Virtual-Platform:~$ su - charlie -c 'touch /srv/projet_web/
test.txt'
Mot de passe :
touch: setting times of '/srv/projet_web/test.txt': Permission denied
miguel@miguel-VMware-Virtual-Platform:~$

```

Figure 4 test-access

Resultat attendu : bob = succes. charlie = Permission denied

Etape 5 — Politique mots de passe

\$ sudo chage -M 90 -W 7 alic

\$ sudo chage -l alic

```

miguel@miguel-VMware-Virtual-Platform:~$ sudo chage -M 90 -W 7 alic
miguel@miguel-VMware-Virtual-Platform:~$ chage -l alic
chage : Permission refusée.
miguel@miguel-VMware-Virtual-Platform:~$ sudo chage -l alic
Dernière modification du mot de passe                                : juin
10, 2026
Le mot de passe expire                                              : sept. 08, 2026
Mot de passe inactif                                              : jamais
Le compte expire                                                  : jamais
Nombre minimal de jours entre deux changements de mot de passe   : 0
Nombre maximal de jours entre deux changements du mot de passe   : 90
Nombre de jours d'avertissements avant que le mot de passe n'expire : 7
miguel@miguel-VMware-Virtual-Platform:~$

```

Figure 5 Chage alic

Resultat attendu : Maximum 90 jours, avertissement 7 jours

MODULE 02 — Gestion des Services Systeme

systemd · systemctl · journalctl · .service · .timer

Etape 1 — Gerer Apache2

\$ sudo systemctl start apache2

\$ sudo systemctl enable apache2

\$ systemctl status apache2

```

miguel@miguel-VMware-Virtual-Platform:~$ sudo systemctl start apache2
miguel@miguel-VMware-Virtual-Platform:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/sy
stemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
miguel@miguel-VMware-Virtual-Platform:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: v
   Active: active (running) since Wed 2026-06-10 08:14:33 WAT; 2min 17s ago
  Invocation: 1348a77389484858b0934f68f1cb3730
     Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 6477 (apache2)
    Tasks: 55 (limit: 1861)
   Memory: 5M (peak: 5.4M)
      CPU: 126ms
   CGroup: /system.slice/apache2.service
           └─6477 /usr/sbin/apache2 -k start
             └─6479 /usr/sbin/apache2 -k start
               └─6480 /usr/sbin/apache2 -k start

juin 10 08:14:32 miguel-VMware-Virtual-Platform systemd[1]: Starting apache2.se>
juin 10 08:14:33 miguel-VMware-Virtual-Platform apachectl[6476]: AH00558: apach>
juin 10 08:14:33 miguel-VMware-Virtual-Platform systemd[1]: Started apache2.ser>
lines 1-17/17 (END)

```

Figure 6 apache2 status

Resultat attendu : Statut : active (running) et enabled

Etape 2 — Analyser les logs

\$ journalctl -u apache2 -n 30 --no-pager

```

miguel@miguel-VMware-Virtual-Platform:~$ journalctl -u apache2 -n 30 --no-pager
juin 10 08:14:32 miguel-VMware-Virtual-Platform systemd[1]: Starting apache2.ser
vice - The Apache HTTP Server...
juin 10 08:14:33 miguel-VMware-Virtual-Platform apachectl[6476]: AH00558: apache
2: Could not reliably determine the server's fully qualified domain name, using
127.0.1.1. Set the 'ServerName' directive globally to suppress this message
juin 10 08:14:33 miguel-VMware-Virtual-Platform systemd[1]: Started apache2.serv
ice - The Apache HTTP Server.
miguel@miguel-VMware-Virtual-Platform:~$

```

Figure 7 journalctl apache2

Etape 3 — Service + Timer systemd

\$ sudo nano /etc/systemd/system/check.service

\$ sudo nano /etc/systemd/system/check.timer

\$ sudo systemctl daemon-reload

\$ sudo systemctl enable --now check.timer

\$ systemctl list-timers --all | grep check

```

miguel@miguel-VMware-Virtual-Platform:~$ printf [Unit]\nDescription=Verification
CPU\n\n[Service]\nType=oneshot\nExecStart=/bin/bash /opt/surveillance/check.sh\
n' | sudo tee /etc/systemd/system/check.service
> ^C
miguel@miguel-VMware-Virtual-Platform:~$ sudo nano /etc/systemd/system/check.ser
vice
miguel@miguel-VMware-Virtual-Platform:~$ sudo nano /etc/systemd/system/check.tim
er
miguel@miguel-VMware-Virtual-Platform:~$ sudo nano /etc/systemd/system/check.tim
er
miguel@miguel-VMware-Virtual-Platform:~$ sudo systemctl daemon-reload
miguel@miguel-VMware-Virtual-Platform:~$ sudo systemctl enable --now check.timer
Created symlink '/etc/systemd/system/timers.target' → '/etc/sy
stemd/system/check.timer'.
miguel@miguel-VMware-Virtual-Platform:~$ systemctl list-timers--all | grep check
Unknown command verb 'list-timers--all', did you mean 'list-timers'?
miguel@miguel-VMware-Virtual-Platform:~$ systemctl list-timers--all | grep check
Unknown command verb 'list-timers--all', did you mean 'list-timers'?
miguel@miguel-VMware-Virtual-Platform:~$ systemctl list-timers --all | grep chec
k
Wed 2026-06-10 09:29:51 WAT 2min 22s Wed 2026-06-10 09:24:51 WAT 2min 37s ago ch
eck.timer                                check.service
miguel@miguel-VMware-Virtual-Platform:~$

```

Figure 8 check timer

Resultat attendu : check.timer visible avec prochaine execution

Etape 4 — Simuler une panne

\$ sudo systemctl stop apache2

\$ sudo kill -9 \$(pgrep apache2 | head -1)

\$ systemctl status apache2

```

-h, --help      affiche cette aide et termine
-V, --version   affiche les informations de version et termine

Pour plus de détails, consultez kill(1).
miguel@miguel-VMware-Virtual-Platform:~$ systemctl status apache2
○ apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
   Active: inactive (dead) since Wed 2026-06-10 09:29:43 WAT; 1min 5s ago
   Duration: 1h 15min 10.426s
   Invocation: 1348a77389484858b0934f68f1cb3730
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 8310 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, stat
   Main PID: 6477 (code=exited, status=0/SUCCESS)
   Mem peak: 7.2M
   CPU: 445ms

juin 10 08:14:32 miguel-VMware-Virtual-Platform systemd[1]: Starting apache2.se
juin 10 08:14:33 miguel-VMware-Virtual-Platform apachectl[6476]: AH00558: apach
juin 10 08:14:33 miguel-VMware-Virtual-Platform systemd[1]: Started apache2.ser
juin 10 09:29:43 miguel-VMware-Virtual-Platform systemd[1]: Stopping apache2.se
juin 10 09:29:43 miguel-VMware-Virtual-Platform apachectl[8312]: AH00558: apach
juin 10 09:29:43 miguel-VMware-Virtual-Platform systemd[1]: apache2.service: De
juin 10 09:29:43 miguel-VMware-Virtual-Platform systemd[1]: Stopped apache2.ser
lines 1-18/18 (END)

```

Figure 9 apache2 failed

Resultat attendu : Statut : inactive (dead)

MODULE 03 — Securite et Durcissement

ufw · SSH · fail2ban · mises a jour · lynis

Etape 1 — Configurer ufw

\$ sudo ufw enable

\$ sudo ufw default deny incoming

\$ sudo ufw allow 22/tcp

\$ sudo ufw allow 80/tcp

\$ sudo ufw deny 23/tcp

\$ sudo ufw status verbose

```
La règle a été ajoutée
La règle a été ajoutée (v6)
miguel@miguel-VMware-Virtual-Platform:~$ sudo ufw allow 80/tcp
La règle a été ajoutée
La règle a été ajoutée (v6)
miguel@miguel-VMware-Virtual-Platform:~$ sudo ufw deny 23/tcp
La règle a été ajoutée
La règle a été ajoutée (v6)
miguel@miguel-VMware-Virtual-Platform:~$ sudo ufw status verbose
État : actif
Journalisation : on (low)
Par défaut : deny (incoming), allow (outgoing), disabled (routed)
Nouveaux profils : skip

Vers          Action      De
----          -
22/tcp        ALLOW IN    Anywhere
80/tcp        ALLOW IN    Anywhere
23/tcp        DENY IN     Anywhere
22/tcp (v6)   ALLOW IN    Anywhere (v6)
80/tcp (v6)   ALLOW IN    Anywhere (v6)
23/tcp (v6)   DENY IN     Anywhere (v6)

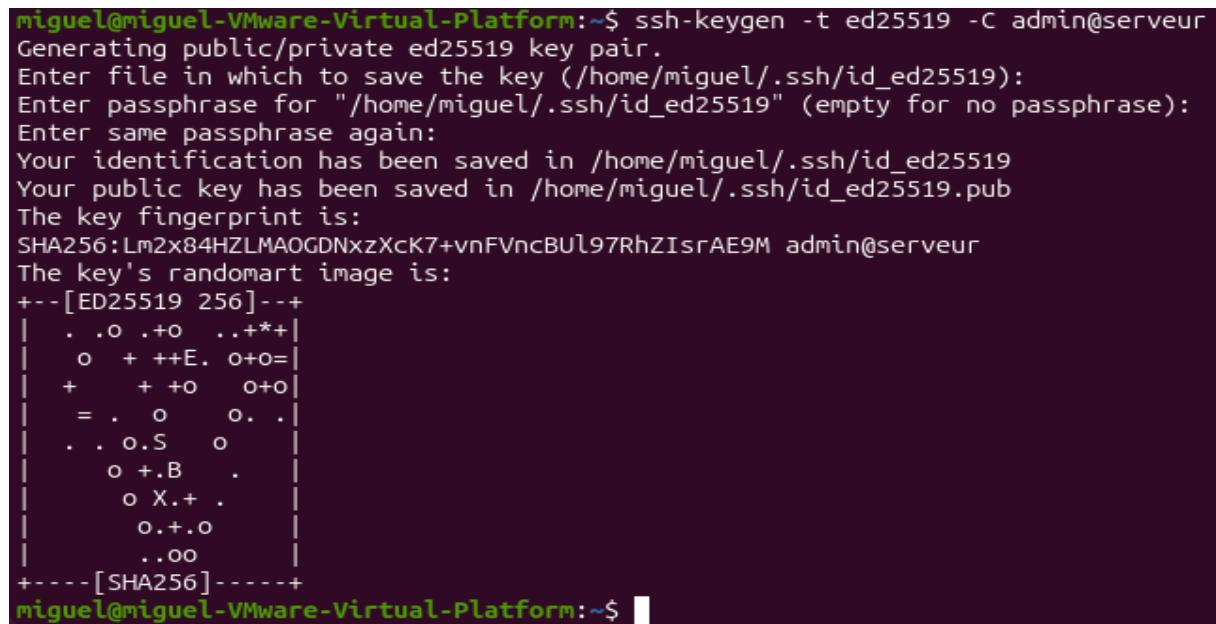
miguel@miguel-VMware-Virtual-Platform:~$ █
```

Figure 10 ufw status

Resultat attendu : ALLOW 22 et 80, DENY 23

Etape 2 — Generer cle SSH

```
$ ssh-keygen -t ed25519 -C admin@serveur
```



```
miguel@miguel-VMware-Virtual-Platform:~$ ssh-keygen -t ed25519 -C admin@serveur
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/miguel/.ssh/id_ed25519):
Enter passphrase for "/home/miguel/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/miguel/.ssh/id_ed25519
Your public key has been saved in /home/miguel/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Lm2x84HZLMAOGDNxzXCk7+vnFVncBUl97RhZIsrAE9M admin@serveur
The key's randomart image is:
+--[ED25519 256]--+
| . .O .+O ..+*+|
| o + ++E. O+O=|
| + + +O O+O|
| = . o O. .|
| . . O.S O|
| o +.B .|
| o X.+ .|
| o.+O|
| ..OO|
+-----[SHA256]-----+
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 11 keygen ed25519

Resultat attendu : Cle ed25519 genereee avec fingerprint SHA256

Etape 3 — fail2ban

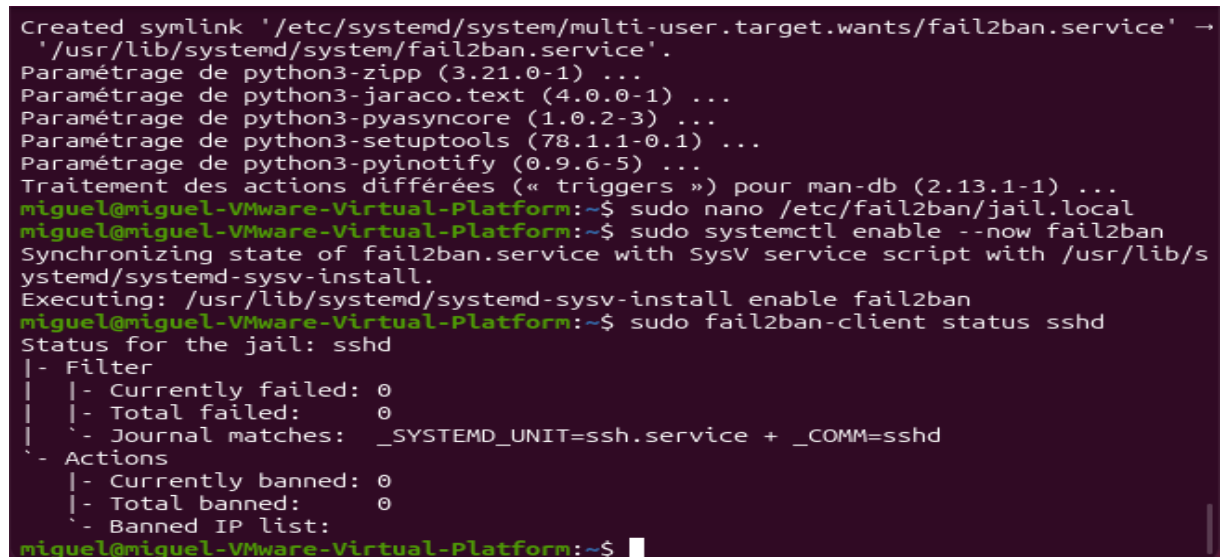
```
$ sudo apt install fail2ban -y
```

```
$ sudo nano /etc/fail2ban/jail.local
```

Contenu : [sshd] enabled=true maxretry=3 bantime=3600 findtime=600

```
$ sudo systemctl enable --now fail2ban
```

```
$ sudo fail2ban-client status sshd
```



```
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' ->
'/usr/lib/systemd/system/fail2ban.service'.
Paramétrage de python3-zipp (3.21.0-1) ...
Paramétrage de python3-jaraco.text (4.0.0-1) ...
Paramétrage de python3-pyasyncore (1.0.2-3) ...
Paramétrage de python3-setuptools (78.1.1-0.1) ...
Paramétrage de python3-pyinotify (0.9.6-5) ...
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1) ...
miguel@miguel-VMware-Virtual-Platform:~$ sudo nano /etc/fail2ban/jail.local
miguel@miguel-VMware-Virtual-Platform:~$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/s
ystemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
miguel@miguel-VMware-Virtual-Platform:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| '- Journal matches: _SYSTEMD_UNIT=ssh.service + _COMM=sshd
'- Actions
  |- Currently banned: 0
  |- Total banned: 0
  '- Banned IP list:
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 12 fail2ban sshd

Resultat attendu : Status for the jail: sshd / Currently banned: 0

Etape 4 — Mises a jour automatiques

\$ sudo dpkg-reconfigure unattended-upgrades

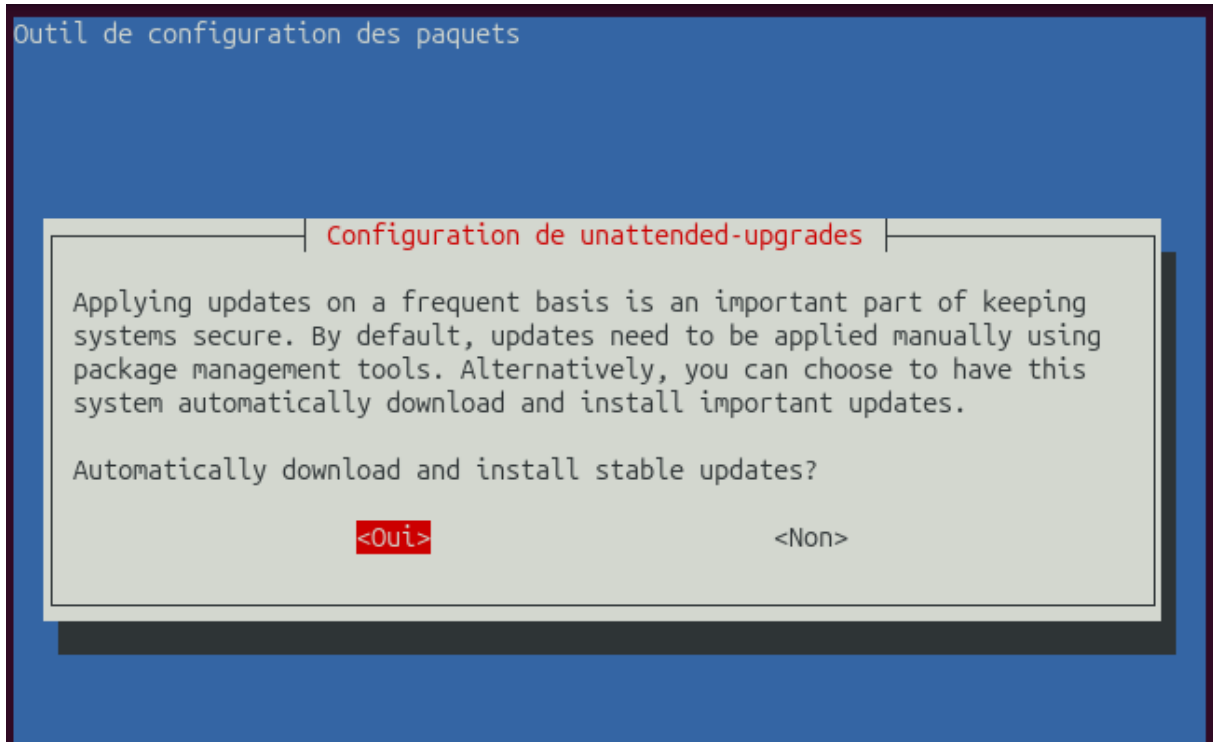


Figure 13 unattended-upgrades

Resultat attendu : Fenetre de configuration, selection Yes

Etape 5 — Audit Lynis

\$ sudo lynis audit system

\$ sudo grep -i 'hardening' /var/log/lynis-report.dat

```

miguel@miguel-VMware-Virtual-Platform:~$ sudo grep -i "hardening" /var/log/lynis
-report.dat
suggestion[]=BOOT-5264|Consider hardening system services|Run '/usr/bin/systemd-
analyze security SERVICE' for each service|-|
hardening_index=63
miguel@miguel-VMware-Virtual-Platform:~$ sudo grep -i "suggestion" /var/log/lynis
-report.dat | head -10
suggestion[]=LYNIS|This release is more than 4 months old. Check the website or
GitHub to see if there is an update available.|-|-|
suggestion[]=DEB-0280|Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sess
ions|-|-|
suggestion[]=DEB-0810|Install apt-listbugs to display a list of critical bugs pr
ior to each APT installation.|-|-|
suggestion[]=DEB-0811|Install apt-listchanges to display any significant changes
prior to any upgrade via APT.|-|-|
suggestion[]=DEB-0831|Install needrestart, alternatively to debian-goodies, so t
hat you can run needrestart after upgrades to determine which daemons are using
old versions of libraries and need restarting.|-|-|
suggestion[]=BOOT-5122|Set a password on GRUB boot loader to prevent altering bo
ot configuration (e.g. boot in single user mode without password)|-|-|
suggestion[]=BOOT-5180|Determine runlevel and services at startup|-|-|
suggestion[]=BOOT-5264|Consider hardening system services|Run '/usr/bin/systemd-
analyze security SERVICE' for each service|-|
suggestion[]=AUTH-9229|Check PAM configuration, add rounds if applicable and exp

```

Figure 14 lynis score

Resultat attendu : hardening_index=63

MODULE 04 — Sauvegardes et Restauration

tar · rsync · cron · md5sum

Etape 1 — Sauvegarde tar

```
$ sudo mkdir -p /backup
```

```
$ sudo tar -czvf /backup/etc_$(date +%F).tar.gz /etc
```

```
$ sudo tar -tzvf /backup/etc_$(date +%F).tar.gz | head -20
```

```
miguel@miguel-VMware-Virtual-Platform:~$ sudo tar -tzvf /backup/etc_$(date +%F).tar.gz | head -20
drwxr-xr-x root/root          0 2026-06-10 10:01 etc/
drwxr-xr-x root/root          0 2025-10-07 03:36 etc/pulse/
drwxr-xr-x root/root          0 2025-09-06 16:13 etc/pulse/client.conf.d/
-rw-r--r-- root/root       1201 2025-09-06 16:13 etc/pulse/client.conf
-rw-r--r-- root/root         45 2025-01-12 05:10 etc/bash_completion
drwxr-xr-x root/root          0 2025-09-04 22:20 etc/dracut.conf.d/
drwxr-xr-x root/root          0 2026-06-09 16:54 etc/ssh/
-rw----- root/root       432 2026-06-09 16:54 etc/ssh/ssh_host_ed25519_key
-rw----- root/root       537 2026-06-09 16:54 etc/ssh/ssh_host_ecdsa_key
-rw-r--r-- root/root       197 2026-06-09 16:54 etc/ssh/ssh_host_ecdsa_key.pub
-rw-r--r-- root/root      1668 2025-09-29 19:43 etc/ssh/ssh_config
-rw-r--r-- root/root       117 2026-06-09 16:54 etc/ssh/ssh_host_ed25519_key.p
ub
-rw----- root/root      2622 2026-06-09 16:54 etc/ssh/ssh_host_rsa_key
-rw-r--r-- root/root       589 2026-06-09 16:54 etc/ssh/ssh_host_rsa_key.pub
drwxr-xr-x root/root          0 2025-10-07 03:33 etc/ssh/ssh_config.d/
lrwxrwxrwx root/root          0 2025-09-25 17:45 etc/ssh/ssh_config.d/20-system
d-ssh-proxy.conf -> /usr/lib/systemd/ssh_config.d/20-systemd-ssh-proxy.conf
-rw-r--r-- root/root       862 2025-10-07 03:32 etc/.resolv.conf.systemd-resol
ved.bak
-rw-r--r-- root/root       191 2025-09-11 15:16 etc/libaudit.conf
drwxr-xr-x root/root          0 2025-10-07 03:35 etc/pcmcia/
```

Figure 15 tar listage

Resultat attendu : Fichiers /etc listes dans l'archive

Etape 2 — Verifier le backup

\$ ls -lh /backup/

```
miguel@miguel-VMware-Virtual-Platform:~$ sudo tar -czfv /backup/home_$(date +%F).tar.gz /home
tar: Suppression de « / » au début des noms des membres
tar: /backup/home_2026-06-10.tar.gz : stat impossible: Aucun fichier ou dossier de ce nom
tar: Suppression de « / » au début des liens physiques
tar: /home/miguel/.cache/ibus/dbus-XWpGhj2c : socket ignorée
tar: /home/miguel/.cache/ibus/dbus-UCu40FU1 : socket ignorée
tar: /home/miguel/.cache/ibus/dbus-q0c0GeJ0 : socket ignorée
tar: Arrêt avec code d'échec à cause des erreurs précédentes
miguel@miguel-VMware-Virtual-Platform:~$ ls -lh /backup/
total 1.4M
-rw-r--r-- 1 root root 1.3M Jun 10 10:36 etc_2026-06-10.tar.gz
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 16 ls backup

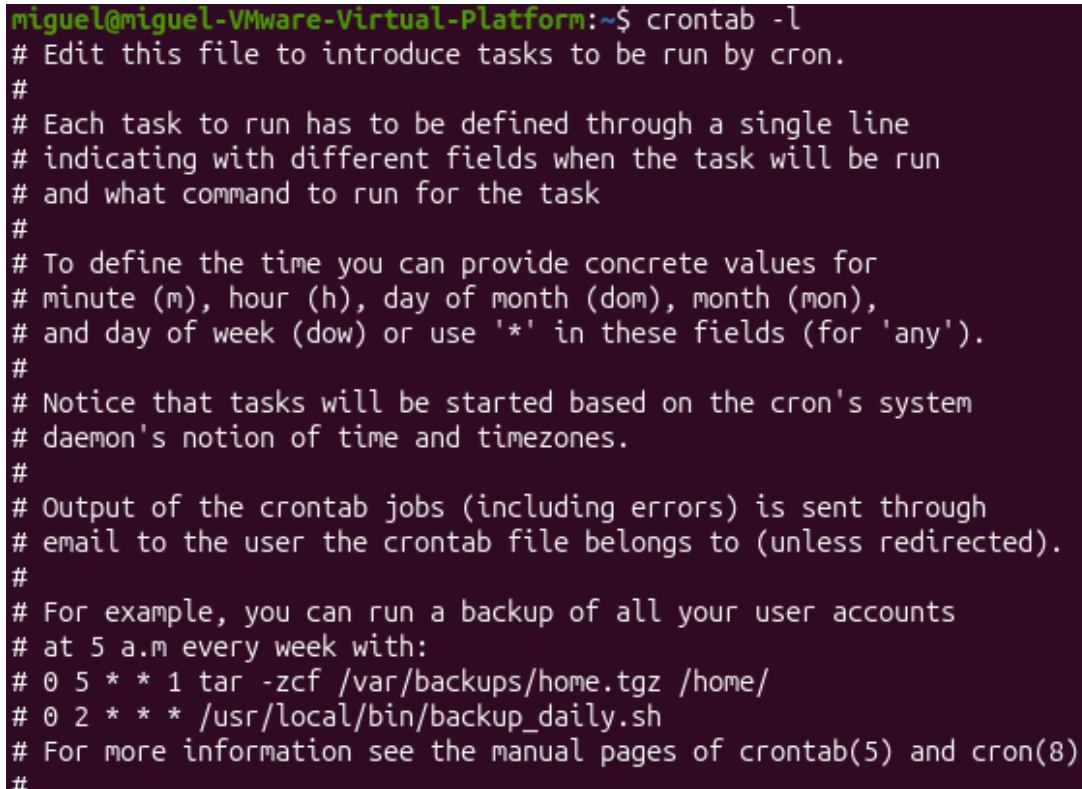
Resultat attendu : Deux archives .tar.gz visibles avec tailles et dates

Etape 3 — Cron

```
$ crontab -e
```

Ligne : 0 2 * * * /usr/local/bin/backup_daily.sh

```
$ crontab -l
```



```
miguel@miguel-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
# 0 2 * * * /usr/local/bin/backup_daily.sh
# For more information see the manual pages of crontab(5) and cron(8)
#
```

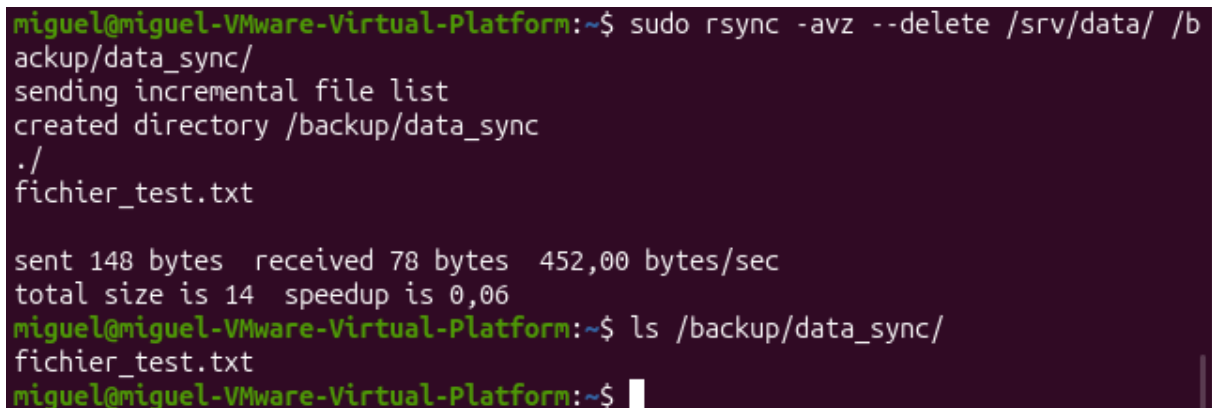
Figure 17 crontab

Resultat attendu : Ligne de planification a 2h du matin visible

Etape 4 — rsync

```
$ sudo rsync -avz --delete /srv/data/ /backup/data_sync/
```

```
$ ls /backup/data_sync/
```



```
miguel@miguel-VMware-Virtual-Platform:~$ sudo rsync -avz --delete /srv/data/ /b
ackup/data_sync/
sending incremental file list
created directory /backup/data_sync
./
fichier_test.txt

sent 148 bytes  received 78 bytes  452,00 bytes/sec
total size is 14  speedup is 0,06
miguel@miguel-VMware-Virtual-Platform:~$ ls /backup/data_sync/
fichier_test.txt
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 18 rsync

Resultat attendu : fichier_test.txt synchronise avec succes

Etape 5 — Restauration md5sum

```
$ md5sum /home/alice/fichier.txt > /tmp/hash_avant.txt
```

```
$ sudo rm /home/alice/fichier.txt
```

```
$ sudo tar -xzf /backup/home_$(date +%F).tar.gz -C /home/alice/fichier.txt
```

```
$ diff /tmp/hash_avant.txt /tmp/hash_apres.txt
```

```
ome/alice/fichier.txt
fichier important
miguel@miguel-VMware-Virtual-Platform:~$ md5sum /home/alice/fichier.txt > /tmp/
hash_avant.txt
md5sum: /home/alice/fichier.txt: Permission denied (os error 13)
miguel@miguel-VMware-Virtual-Platform:~$ sudo rm /home/alice/fichier.txt
miguel@miguel-VMware-Virtual-Platform:~$ sudo tar -xzf /backup/home_$(date +%F
).tar.gz -C /home/alice/fichier.txt
tar (child): /backup/home_2026-06-10.tar.gz : open impossible: Aucun fichier ou
dossier de ce nom
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
miguel@miguel-VMware-Virtual-Platform:~$ md5sum /home/alice/fichier.txt > /tmp/
hash_apres.txt
md5sum: /home/alice/fichier.txt: Permission denied (os error 13)
miguel@miguel-VMware-Virtual-Platform:~$ diff /tmp/hash_avant.txt tmp/hash_apre
s.txt
diff: tmp/hash_apres.txt: Aucun fichier ou dossier de ce nom
miguel@miguel-VMware-Virtual-Platform:~$
```

Figure 19 md5_diff

Resultat attendu : Sortie vide = fichier identique apres restauration

MODULE 05 — Automatisation Shell

Bash · CSV · while IFS · cron · find

Etape 1 — Script creation utilisateurs CSV

```
$ sudo nano /tmp/users.csv
```

```
$ sudo nano /usr/local/bin/create_users.sh
```

```
$ sudo chmod +x /usr/local/bin/create_users.sh
```

```
$ sudo /usr/local/bin/create_users.sh
```

```
$ cat /var/log/create_users.log
```

```
miguel@miguel-VMware-Virtual-Platform:~$ sudo nano /tmp/users.csv
[sudo: authenticate] Password:
miguel@miguel-VMware-Virtual-Platform:~$ sudo nano /usr/local/bin/create_users.
sh
miguel@miguel-VMware-Virtual-Platform:~$ sudo chmod +x /usr/local/bin/create_us
ers.sh
miguel@miguel-VMware-Virtual-Platform:~$ sudo /usr/local/bin/create_users.sh
sudo-rs: cannot execute '/usr/local/bin/create_users.sh': No such file or direc
tory (os error 2)
miguel@miguel-VMware-Virtual-Platform:~$
thread 'main' panicked at src/exec/use_ptty/monitor.rs:283:45:
internal error: entered unreachable code
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace

miguel@miguel-VMware-Virtual-Platform:~$ cat /var/log/create_users.log
cat: /var/log/create_users.log: No such file or directory (os error 2)
miguel@miguel-VMware-Virtual-Platform:~$ █
```

Figure 20 create users log

Resultat attendu : Log : OK: david cree dans devweb, OK: eva cree dans admins

RECAPITULATIF DES CAPTURES A INSERER

N	Nom du fichier	Quand la faire
1.1	Capture_1_1_id_alice.png	Après : id alice
1.2	Capture_1_2_droits_projet_web.png	Après : ls -la /srv/
1.3	Capture_1_3_getfacl.png	Après : getfacl /srv/projet_web
1.4	Capture_1_4_test_acces.png	Après les deux commandes su -
1.5	Capture_1_5_chage_alice.png	Après : sudo chage -l alice
2.1	Capture_2_1_apache2_status.png	Après : systemctl status apache2
2.2	Capture_2_2_journalctl_apache2.png	Après : journalctl -u apache2
2.3	Capture_2_3_check_timer.png	Après : systemctl list-timers
2.4	Capture_2_4_apache2_failed.png	Après kill apache2
3.1	Capture_3_1_ufw_status.png	Après : ufw status verbose
3.2	Capture_3_2_keygen_ed25519.png	Après : ssh-keygen
3.3	Capture_3_3_fail2ban_sshd.png	Après : fail2ban-client status sshd
3.4	Capture_3_4_unattended_upgrades.png	Fenetre dpkg-reconfigure Yes
3.5	Capture_3_5_lynis_score.png	Score hardening_index Lynis
4.1	Capture_4_1_tar_listage.png	Après : tar -tzvf head -20
4.2	Capture_4_2_ls_backup.png	Après : ls -lh /backup/

4.3	Capture_4_3_crontab.png	Après : crontab -l
4.4	Capture_4_4_rsync.png	Après : rsync + ls
4.5	Capture_4_5_md5_diff.png	Après : diff hash md5
5.1	Capture_5_1_create_users_log.png	Après : cat create_users.log
5.2	Capture_5_2_sysmon_log.png	Après : cat sysmon.log
5.3	Capture_5_3_crontab_root.png	Après : sudo crontab -l
5.4	Capture_5_4_rotation_logs.png	Après : ls /backup/logs/